

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Parallel Group, Placebo-controlled Multicenter Study to Evaluate Efficacy, Safety and Tolerability of Ianalumab in Participants With Diffuse Cutaneous Systemic Sclerosis **Short Title/Acronym:** A Clinical Study to Evaluate Ianalumab in Participants With Diffuse Cutaneous Systemic Sclerosis (VENUSS) **Protocol Number:** CVAY736S12201 / 2024-511933-36-00 **NCT Number:** NCT06470048 **Sponsor Name:** Novartis Pharma AG **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of ianalumab compared to placebo in improving disease activity and reducing disease progression in participants with diffuse cutaneous systemic sclerosis (dcSSc) at Week 52. **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of subcutaneous ianalumab, and to assess its effects on skin thickness (modified Rodnan skin score) and lung function (forced vital capacity).

2.2 Background & Rationale Diffuse cutaneous systemic sclerosis (dcSSc) is a severe autoimmune disease characterized by immune dysregulation, vascular dysfunction, and progressive fibrosis of the skin and internal organs. There is a high unmet medical need for disease-modifying therapies. Ianalumab is a novel, fully human monoclonal antibody that targets B cells via two mechanisms: antibody-dependent cellular cytotoxicity (ADCC) and blockade of the BAFF receptor. By depleting B cells and interrupting survival signals, ianalumab has the potential to alter the autoimmune and fibrotic pathways driving dcSSc.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple: Participant, Care Provider, Investigator, Outcomes Assessor) **Randomization:** Randomized (Parallel Group)

3.2 Planned Enrollment & Duration Target **SN\$:** 200 participants **Number of Sites:** Multicenter (Global, including sites in the US, Europe, and Japan) **Estimated Study Start:** 08-Oct-2024 **Estimated Primary Completion:** 14-Jul-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants ≥ 18 and ≤ 70 years of age at the time of the screening visit. 2. Diagnosis of systemic sclerosis per the 2013 ACR/EULAR classification criteria, meeting the dcSSc subset classification according to LeRoy. 3. Disease duration of ≤ 60 months. 4. Positive for relevant autoantibodies (e.g., anti-topoisomerase I [ATA], anti-RNA polymerase III, or ANA $\geq 1:80$ limited to 30% of the cohort).

4.2 Exclusion Criteria 1. Rheumatic disease other than dcSSc (e.g., limited cutaneous disease [lcSSc] or sine scleroderma). 2. Treatment with biologic agents, intravenous immunoglobulin, or cyclophosphamide within 12 weeks prior to baseline. 3. Prior use of B-cell depleting therapies (e.g., rituximab) within 36 weeks prior to randomization. 4. Use of anti-fibrotic agents (e.g., nintedanib, pirfenidone, colchicine, D-penicillamine) within 4 weeks prior to the baseline visit.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ianalumab versus matching placebo administered at specified intervals. **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 52 weeks of double-blind treatment (Treatment Period 1), followed by 52 weeks of open-label ianalumab treatment (Treatment Period 2).

5.2 Concomitant & Prohibited Therapy Permitted: Standard background therapies for SSc symptom management not explicitly prohibited. **Prohibited:** Concurrent investigational agents (within 4 weeks of baseline), biologics, B-cell depleters, and anti-fibrotic agents for ILD management (unless permitted by protocol exceptions).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 6 weeks	Informed Consent, Eligibility Confirmation, Autoantibody/Lab panels, Baseline mRSS
2	Weeks 0 to 52	Randomization, Double-blind s.c. injections (Ianalumab/Placebo), Safety monitoring, Efficacy Assessments
3	Weeks 52 to 104	Open-label ianalumab s.c. injections, Efficacy, and Safety follow-up
4	20 weeks to 2 years	Post-Treatment Follow-Up
5		Final safety evaluations, B-cell reconstitution monitoring, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 3/5 Revised Composite Response Index in Systemic Sclerosis 25 (rCRISS25) response at Week 52.

Measurement Method: Composite clinical assessment measuring systemic sclerosis disease activity and progression.

7.2 Secondary & Exploratory Endpoints 1. Change in modified Rodnan skin score (mRSS) from baseline. 2. Change from baseline in forced vital capacity (FVC). 3. Pharmacokinetics, safety, and tolerability of ianalumab.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through investigator evaluations, lab tests, and patient reports for the duration of the study and follow-up. **Serious Adverse Events (SAEs):** Must be reported within 24 hours to the sponsor and Data Safety and Monitoring Board (DSMB). **Data Monitoring Committee (DMC):** An independent DSMB is appointed to oversee the trial and review unblinded safety data at regular intervals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target enrollment of $N=200$ participants provides sufficient statistical power to detect a clinically meaningful difference in the rCRISS25 response rate at Week 52. **Handling of Missing Data:** Advanced imputation techniques for dropouts or missing continuous measures (e.g., mRSS, FVC), per the final SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine remote and onsite monitoring visits by the sponsor/CRO. **Audit Trail:** Fully validated Electronic Data Capture (EDC) system with secure audit trails and standardized eCRF protocols.

11. Study Identifiers & Governance

Primary ID: CVAY736S12201 **Registry Number:** NCT06470048 **Sponsor/Lead Organization:** Novartis Pharma AG **Study Title:** A Clinical Study to Evaluate Ianalumab in Participants With Diffuse Cutaneous Systemic Sclerosis **Official Regulatory Title:** A

Randomized, Double-blind, Parallel Group, Placebo-controlled Multicenter Study to Evaluate Efficacy, Safety and Tolerability of Ianalumab in Participants With Diffuse Cutaneous Systemic Sclerosis

12. Clinical Framework (SSc Specific)

Primary Condition: Diffuse Cutaneous Systemic Sclerosis (dcSSc)
Diagnosis Threshold: Diagnosis defined by the 2013 ACR/EULAR classification criteria for SSc with dcSSc subset classification.
Medical Methodology: B-cell targeted biologic disease-modifying therapy. **Optimization Target:** Reduction in skin fibrosis (mRSS) and stabilization of organ involvement (FVC/ILD).

13. Study Procedures & Methodology

Management Pathway: Targeted B-cell inhibition using subcutaneous monoclonal antibody. **Education Protocol (HCP):** Site initiation visits and standardized training on performing mRSS and other SSc-specific outcome measures. **Education Protocol (Patient):** Informed consent detailing trial procedures, self-reporting methods, and potential B-cell depletion risks. **Quality Audit Mechanism:** Centralized scoring reviews for mRSS and FVC readouts to minimize inter-rater variability.

14. Observation & Follow-Up Schedule

Total Duration: Up to 104 weeks of treatment plus up to 2 years of post-treatment follow-up. **Onsite Visit Cadence:** Regular clinic visits throughout Treatment Period 1 (52 weeks) and Treatment Period 2 (52 weeks). **Remote Monitoring Frequency:** Intermittent remote check-ins for safety as per site practices. **Checklist Review Interval:** Regular review of concomitant medications and adverse events at each scheduled protocol visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				